⁕ ASSIGNMENT-3 ⁕

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ue.1] Explain with diagram conversion of JK F/F to D F/F.

⇒ Answer :

i) The truth table is as follows :

Inputs			Outputs	
D	Previous state (Q_n)	Next state (Q_{n+1})	J	K
0	0	0	0	X
1	0	1	1	X
0	1	0	X	1
1	1	1	X	0

⟵ Excitation table of D F/F ⟶

⟵ Excitation table of JK F/F ⟶

ii) K-maps & simplification :

For J output

D \ Q_n	0	1
0	0	X
1	1	X

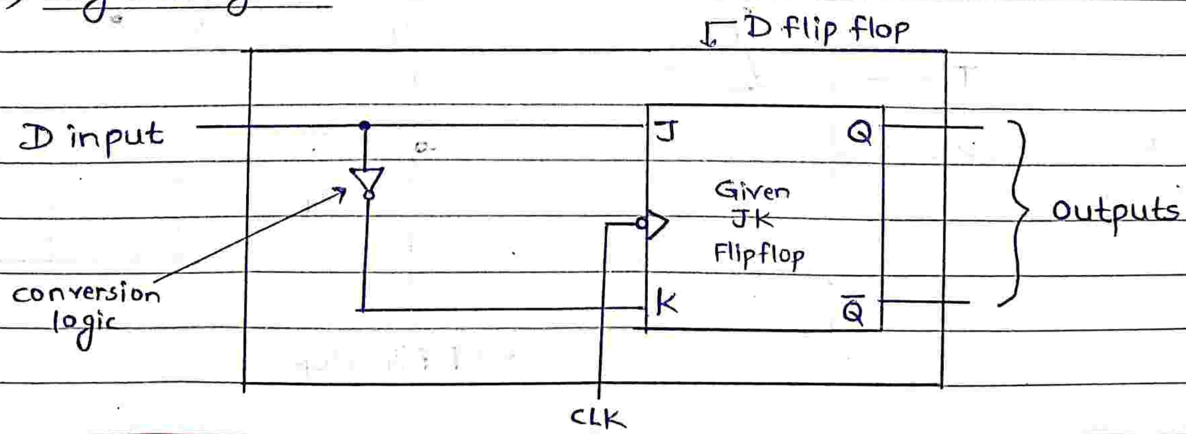
$J = D$

For K output

D \ Q_n	0	1
0	X	1
1	X	0

$K = \overline{D}$

iii) Logic Diagram :



Que.2] Explain with the diagram, how ① D F/F can be converted T flip flop, ⓘⓘ SR to D, ⓘⓘⓘ SR to JK.

⟹ Answer :

① D Flip flop to T Flip Flop :

Inputs			Outputs
T	Present state Q_n	Nextstate Q_{n+1}	D
0	0	0	0
1	0	1	1
1	1	0	0
0	1	1	1

K-map for output D ⟹

T \ Q_n	0	1
0	0	1
1	1	0

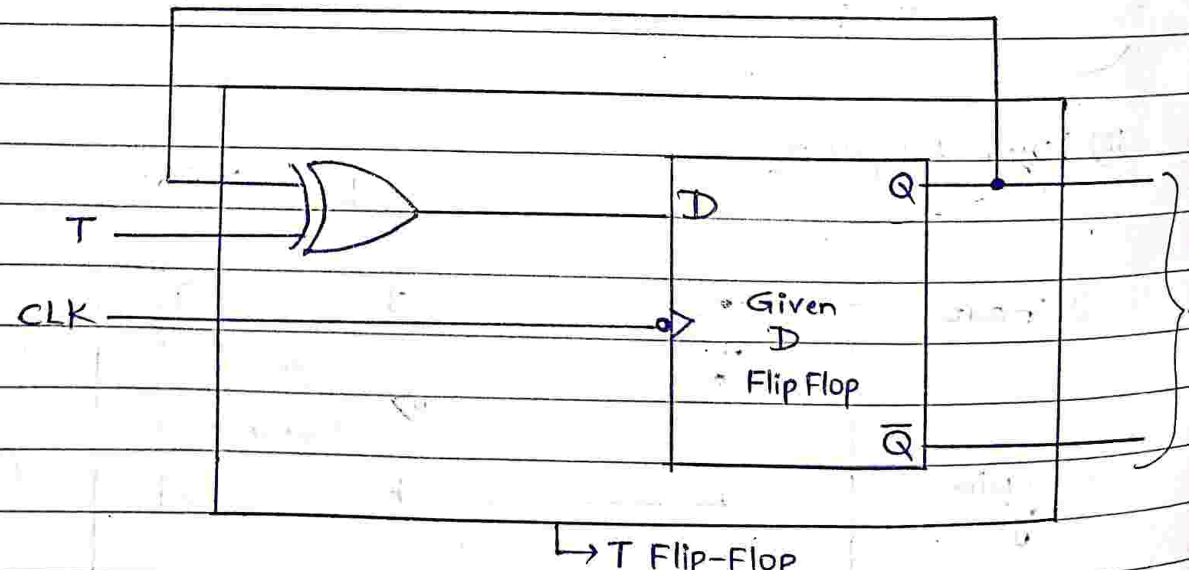
$\rightarrow \overline{T} Q_n$

$\rightarrow T \overline{Q}_n$

$\therefore D = T\overline{Q}_n + \overline{T} Q_n$

$\therefore D = T \oplus Q_n$

Logic Diagram



$\hookrightarrow$ T Flip-Flop

(ii) SR Flip Flop to D Flip Flop:

Inputs			Outputs	
D	Present state Q_n	Nextstate Q_{n+1}	S	R
0	0	0	0	X
1	0	1	1	0
0	1	0	0	1
1	1	1	X	0

K-map for S & R outputs:

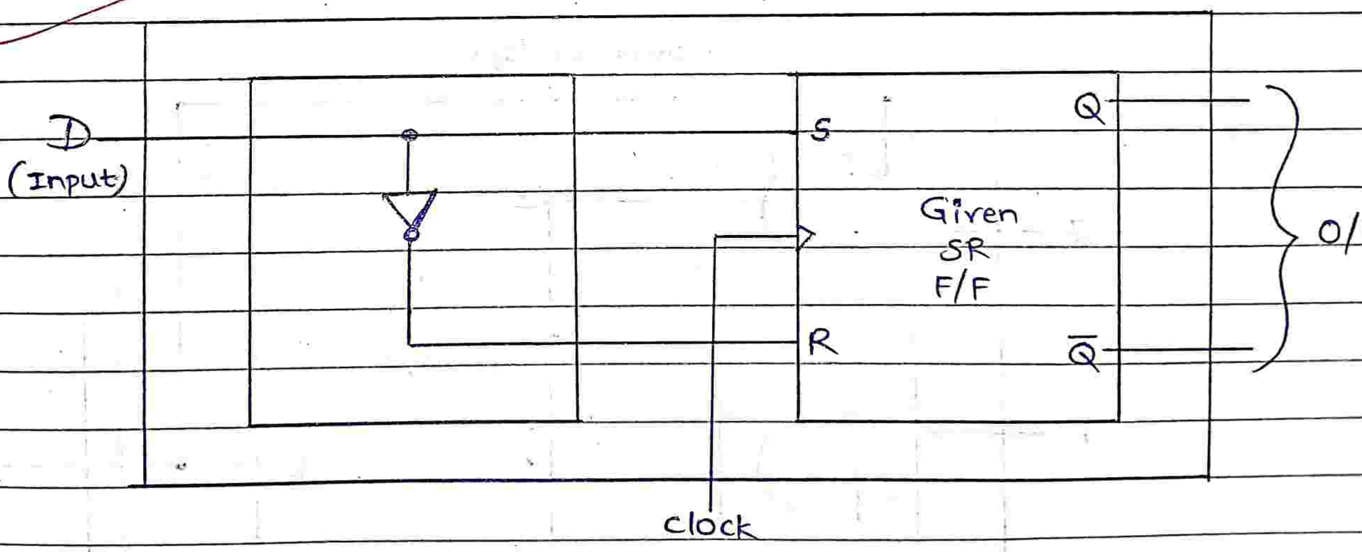
D \ Q_n	0	1
0	0	0
1	1	X

$S = D$

D \ Q_n	0	1
0	X	1
1	0	0

$R = \overline{D}$

Logic Diagram



(iii) SR Flip Flop to JK Flip Flop :-

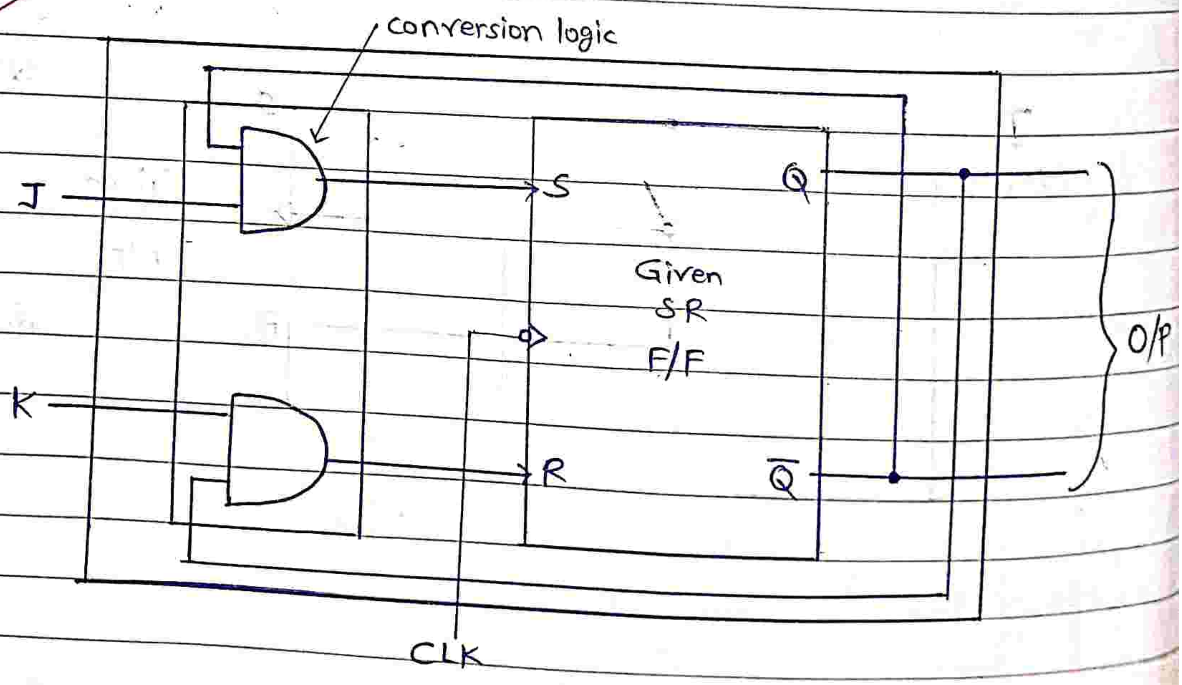
Inputs				Outputs	
J	K	Present state Q_n	Next state Q_{n+1}	S	R
0	0	0	0	0	X
0	1	0	0	0	X
1	0	0	1	1	0
1	1	0	1	1	0
0	1	1	0	0	1
1	1	1	0	0	1
0	0	1	1	X	0
1	0	1	1	X	0

J \ KQ_n	00	01	11	10
0	0	X	0	0
1	1	X	0	1

$S = J\overline{Q}_n$

J \ KQ_n	00	01	11	10
0	X	0	1	X
1	0	0	1	0

$R = KQ_n$



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Que.3] Differentiate between combinational & sequential circuit.

⇒ Answer :

Parameter	combinational circuits	Sequential circuits
1) output depends on	Inputs present at that instant of time.	Present inputs & past inputs/outputs.
2) Memory.	Not Necessary.	Necessary.
3) clock I/P	Not Necessary	Necessary.
4) Examples	Adders, subtractors, code converters.	Flip flops, shift registers, counters.

Que.4] Diff. betn Latch & Flip Flop w.r.t. defination, operation & application.

⇒ Answer :

Parameters	Flip flop	Latch
1) Basic principle	It utilizes an edge triggering approach.	It follows a level triggering approach.
2) working	It works using binary i/p & clock signal.	It operates only using binary i/ps.
3) operating speed	Slow operating speed	Fast operating speed.
4) operations	It performs synchronous operation	It perform Asynchronous operations
5) Applications	They constitute building block of many sequential ckts such as counters.	For designing sequential ckts. (Not generally preferred).

Que. 5] Design 3-bit up/down synchronous counter using MS-JK F/F (IC 7476). Draw logic diagram.

⇒ **Answer :**

i) The number of Flip flops to be used is three.
Let upcounting takes place with M=0 & down counting takes place for M=1.

ii) Excitation Table :-

Mode control (M)	Present state			Next state			F/F Inputs			
	Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	T_C	T_B	T_A	
0	0	0	0	0	0	1	0	0	1	Up counting
0	0	0	1	0	1	0	0	1	1	
0	0	1	0	0	1	1	0	0	1	
0	0	1	1	1	0	0	1	1	1	
0	1	0	0	1	0	1	0	0	1	
0	1	0	1	1	1	0	0	1	1	
0	1	1	0	1	1	1	0	0	1	
0	1	1	1	0	0	0	1	1	1	
1	0	0	0	1	1	1	1	1	1	Down counting
1	0	0	1	0	0	0	0	0	1	
1	0	1	0	0	0	1	0	1	1	
1	0	1	1	0	1	0	1	0	1	
1	1	0	0	0	1	1	0	1	1	
1	1	0	1	1	0	0	0	0	1	
1	1	1	0	1	0	1	0	1	1	
1	1	1	1	1	1	0	0	0	1	

iii) K-maps:

MQ_C \ Q_BQ_A	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$\therefore T_C = \overline{M}Q_BQ_A + M\overline{Q}_B\overline{Q}_A$

MQ_C \ Q_BQ_A	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$\therefore T_B = \overline{M}Q_A + M\overline{Q}_A$

$= M \oplus Q_A$

MQ_C \ Q_BQ_A	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$\therefore T_A = 1$

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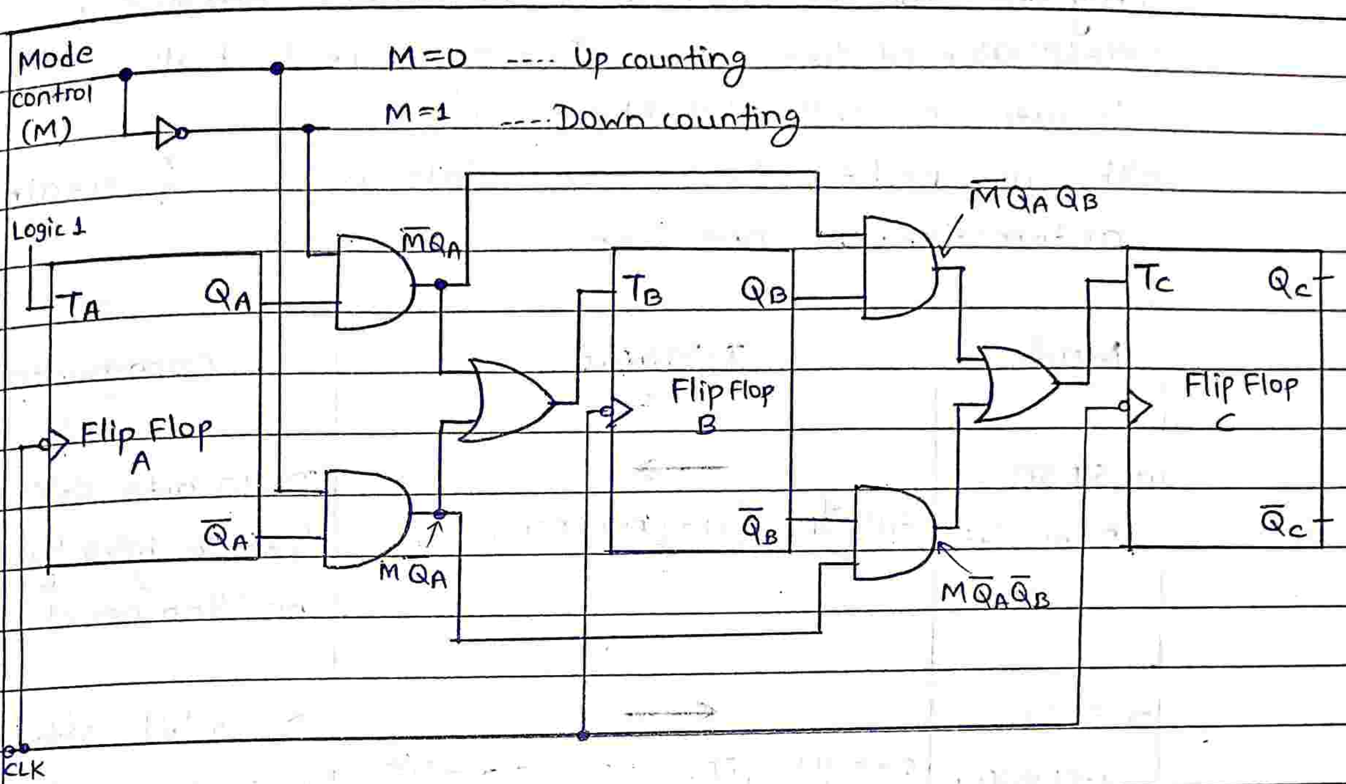
iv) Logic Diagram :



Logic diagram for a 3 bit synchronous Up Down counter

Que.6] What is shift Register ? state the types of shift Registers with application of each.

⟹ Answer :

i) Flip Flops is a 1 bit memory cell which can be used for storing the digital data.

ii) To increase the storage capacity in terms of no. of bits, we have to use a group of flip flops. Such a group of flip flops is known as Register.

iii) Data can be entered in serial or parallel manner into register.

iv) Registers are classified on the basis of how the data bits are entered & taken out from register.

The four possible modes of operation are:

1] Serial In Serial Out (SISO).
2] Serial In Parallel Out (SIPO).
3] Parallel In Serial Out (PISO)
4] Parallel In Parallel Out (PIPO)

v) The binary data (input) in a register can be moved within the register from one flip flop to the other or outside it with application of clock pulses. The registers that allow such data transfers are called 'shift registers'.

vi) They are used for data storage, data transfer & certain arithmetic & logic operations.

Mode	Diagram	Comments
1) SISO (Right shift)	→ I/P → FF3 | FF2 | FF1 | FF0 → O/P	Data bits shift from left to right by 1 position per clock cycle.
2) SISO (Left shift)	← O/P ← FF3 | FF2 | FF1 | FF0 ← I/P	Data bits shift from right to left by 1 position per clock.
3) SIPO	I/P → FF3 | FF2 | FF1 | FF0 ↓ ↓ ↓ ↓ O/Ps	All output bits are made available simultaneously after 4 clock pulses.
4) PISO	I/Ps ↓ ↓ ↓ ↓ FF3 | FF2 | FF1 | FF0 → O/P	All inputs are loaded simultaneously but o/p bit by bit
5) PIPO	Inputs ↓ ↓ ↓ ↓ FF3 | FF2 | FF1 | FF0 ↓ ↓ ↓ ↓ outputs	All inputs are loaded simultaneously & are available at output simultaneously

Que.7] Describe modulus of counter. Design MOD82 counter using Decode counter IC 7490.

⇒ Answer :

i) Modulus (MOD) of a counter represents the number of states through which the counter progresses during its operation. It is denoted by N.

ii) Thus, MOD-N counter mean's the counter progresses thr N states.

iii) In general, m number of flip Flops are required to construct mod-n counter, where $N \leq 2^m$.

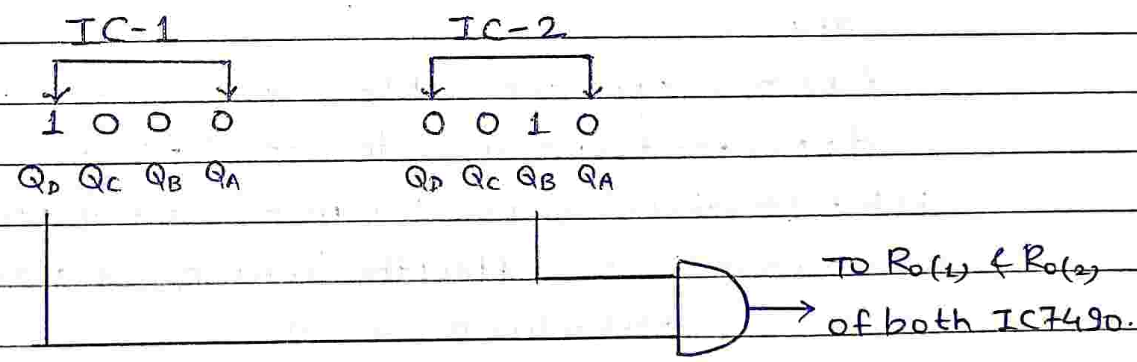
iv) Design MOD82 counter :

Step-1] Number of IC's required :

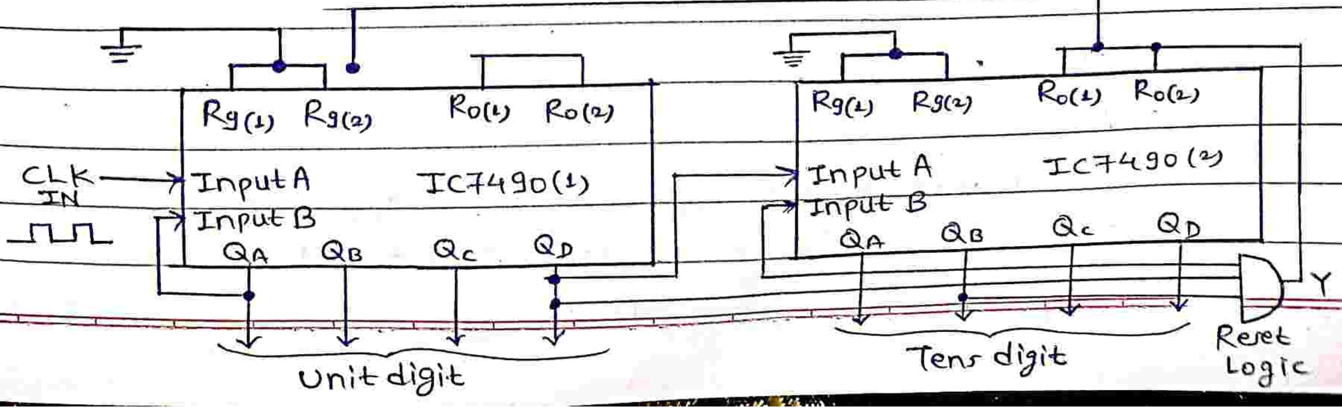
Upto MOD-100, two IC 7490s will be sufficient.

Step-2] Design of Reset Logic :

In order to reset the counter at 82, connect the Q outputs which are equal to 1 in the count of 82 to an AND gate & then connect AND output to $R_{0(1)}$ & $R_{0(2)}$ i.e. reset inputs of both ICs.



Step-3] Logic Diagram :



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Que.1] Explain following terms in brief.

a) ALU signals.

⟹ Ans :

i) ALU contains a variety of electrical input & output connections which result in the digital signals being cast between the ALU & external electronics. External circuits send signals to the ALU input & ALU sends signals to external electronics.

ii) Opcode : The operation selection code specifies whether the ALU will conduct arithmetic or logic operation when it performs the operation.

iii) Data : The ALU contains three parallel buses, each with two input & output operands. These three buses are in charge of same amount of signals.

b) ALU Functions.

⟹ Ans :

i) Some ALUs are designed solely to conduct integer calculations whereas others are built to perform floating-point computations.

ii) Arithmetic operators : It refers to bit subtractors & additions, despite the fact that it does multiplication & division.

iii) Bit-shifting operators : It is responsible for multiplication, which involves shifting the locations of a bit to the right or left by particular no. of places.

iv) Logical operations : These consist of NOR, AND, NOT, NAND, XOR, OR & more.

c) ALU Types

⟹ Ans :

i) Arithmetic Unit & Logic unit are two types of ALU.

ii) ALU consists of three types of functional parts : storage registers, operations logic & sequencing logic.

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Que.2] Write in brief about Fetch cycle with registers involved & sequence of micro-instructions to carry out those operations.

⇒ Ans:

i) Fetch cycle is concerned to fetch (i.e. read from memory) the instruction.

ii) It involves following operations in different t-states (t-state is a time state & is equal to one clock pulse) & hence the mentioned microinstructions below:

	Operation	Microinstructions
T1	PC ⟶ MAR	PC_{out} MAR_{in}, Read, clear y, set C_{in}, Add, Z_{in}
T2	M ⟶ MBR PC ⟵ PC+1	Z_{out}, PC_{in}, wait for memory fetch cycle.
T3	MBR ⟶ IR	MBR_{out}, IR_{in}

iii) As seen in table, three clock pulses or t-states are required for fetch cycle.

iv) control unit is an organizational part of CPU, hence the design can vary from processor to processor.

v) In first t-state, address of instruction to be executed is given to MAR register from PC register.

vi) To perform this operation, control signals given are PC_{out} & MAR_{In}.

vii) This will make PC register give out its data & MAR register accept this data. Also, memory is indicated to perform a read operation hence signal 'Read'.

viii) To increment value of PC, various operations are performed on ALU signals i.e. clear Y, set C_{in}, Add, Z_{in}.

ix) In second clock pulse, CPU has to wait for memory operation, but in same time, it can transfer result in 'Z' register to PC register with control signals namely Z_{out} & PC_{in}.

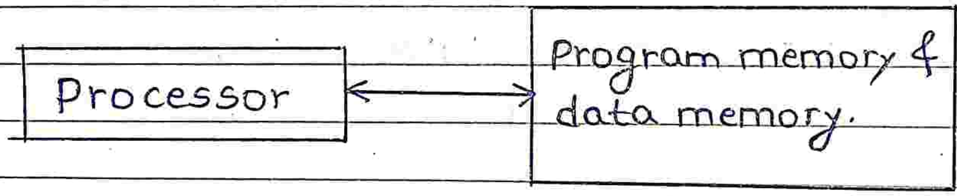
x) This could not be done in previous t-state, as two data cannot be given simultaneously on data bus, else it will get mixed up.

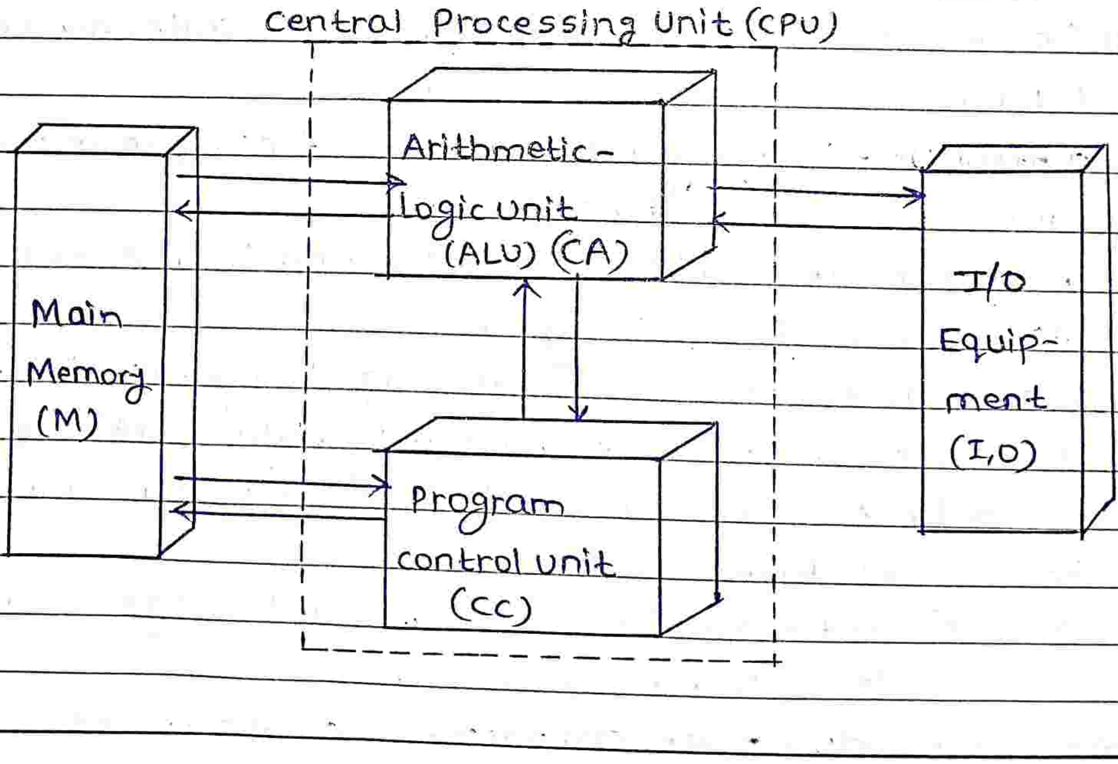
xi) Only one data can be given on the data bus in any clock pulse, but as many as required can accept data.

xii) In final t-state, contents received from memory i.e. instruction is transferred to its correct place. This is done by control signals namely MB_{Rout} & IR_{in}.

Que.3] Explain in brief various function units of computer system with block diagram, showing interconnection between them.

⇒ Ans: (Von-Neumann Architecture).





i) The computer has common memory for data as well as code to be executed.

ii) Processor needs two clock cycles to complete an instruction, first to get an instruction & second to get data.

iii) This system has 3 units : CPU, Memory & I/O devices. CPU has two units : Arithmetic Unit & Control Unit.

iv) Input Unit :-
   - Computer accepts inputs from user thr input devices.
   - e.g. Keyboard, mouse, camera, scanner, joystick, etc

v) Output Unit :-
   - Result is given back by the computer to the user through an output device.
   - I/P & O/P devices are called as human interface devices.
   - e.g. Monitor, Printer, speaker, etc.

vi) Arithmetic & Logic Unit (ALU) :-
   - It performs operations like Addition, Multiplication, division, AND, OR, EXOR, etc.

vii) Control Unit :-
   - It is the main unit that controls all operations of system, inside & outside processor.

viii) Memory Unit :-
   It is used to store programs & data for computer.

Que.4] Write short note on PC, MAR, MBR, IR.

⟹ Ans :

i) PC stands for Program counter. It always contains the address of next instruction to be executed.

ii) MAR stands for Memory Address Register. It provides address of memory location from where data or instruction is to be retrieved or to which data is to be stored.

iii) MAR is also used to forward the address of memory location where data is to be stored.

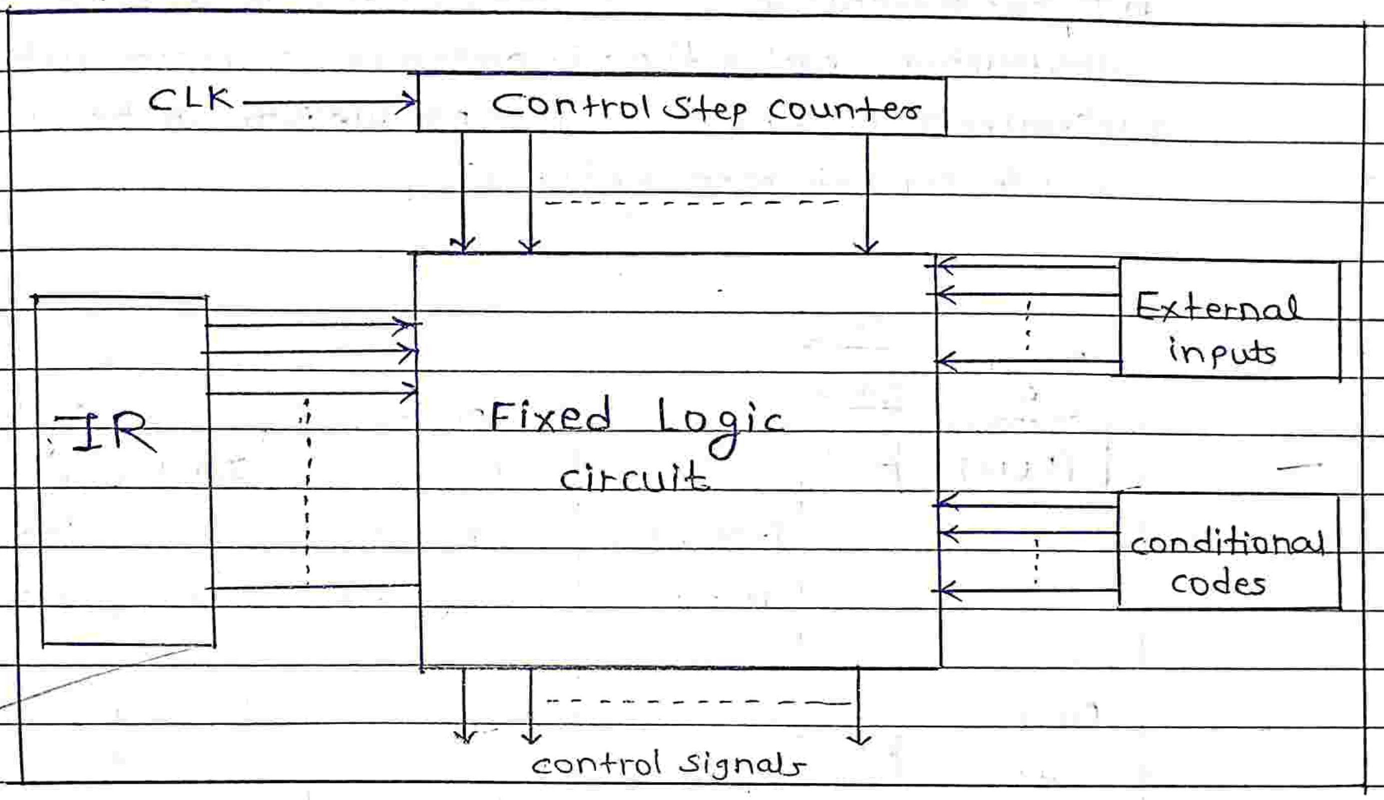
iv) IR i.e. Instruction Register holds the current instruction i.e. opcode & operand of instruction to be executed.
v) MBR i.e. Memory Data/Buffer Register contains the current data fetched from memory.

Que.5] What is function of control unit in CPU ? Draw block diagram of Hardwired control unit & explain its operation, pros & cons.

⟹ Ans :

i) The control unit is the main unit that controls all the operations of system, inside & outside processor.
ii) The memory or I/O devices have to be controlled by the computer to perform operation according to instruction given to it.
iii) The unit which directs the operation of the processor & is a part of CPU is known as control unit. There are 2 types of control units :
   1] Hardwired control unit.
   2] Micro-programmed control unit.
iv) Hardwired control Unit :
   - To interpret the Instructions & generate control signals for them, this control unit uses fixed logic circuits.
   - To generate signals, fixed logic circuits use contents of control step counter, Instruction Register (IR) & code flag & some external input signals such as interrupt signals.
   - The fixed logic ckt in diagram is combinational ckt made from decoders & encoders. It generates the o/p based on state of its inputs.
   - Decoder decodes instruction loaded in IR & generates signal that serves as input to encoder.

- Also, external input & conditional codes act as an input to encoder. Encoder then accordingly generates control signals based on inputs.
- After execution of each instruction, another signal: end signal is generated which resets state of control step counter & makes it ready for next instruction.



v) Advantages:

1] Because of use of combinational ckts to generate signals, Hardwired control unit is fast.
2] It depends on no. of gates, how much delay can occur in generation of control signals.
3] It does not require control memory.

vi) Disadvantages:

1] Complexity of design increases as we require more control signals to be generated.
2] Adding a new feature is difficult & complex.
3] It is expensive.

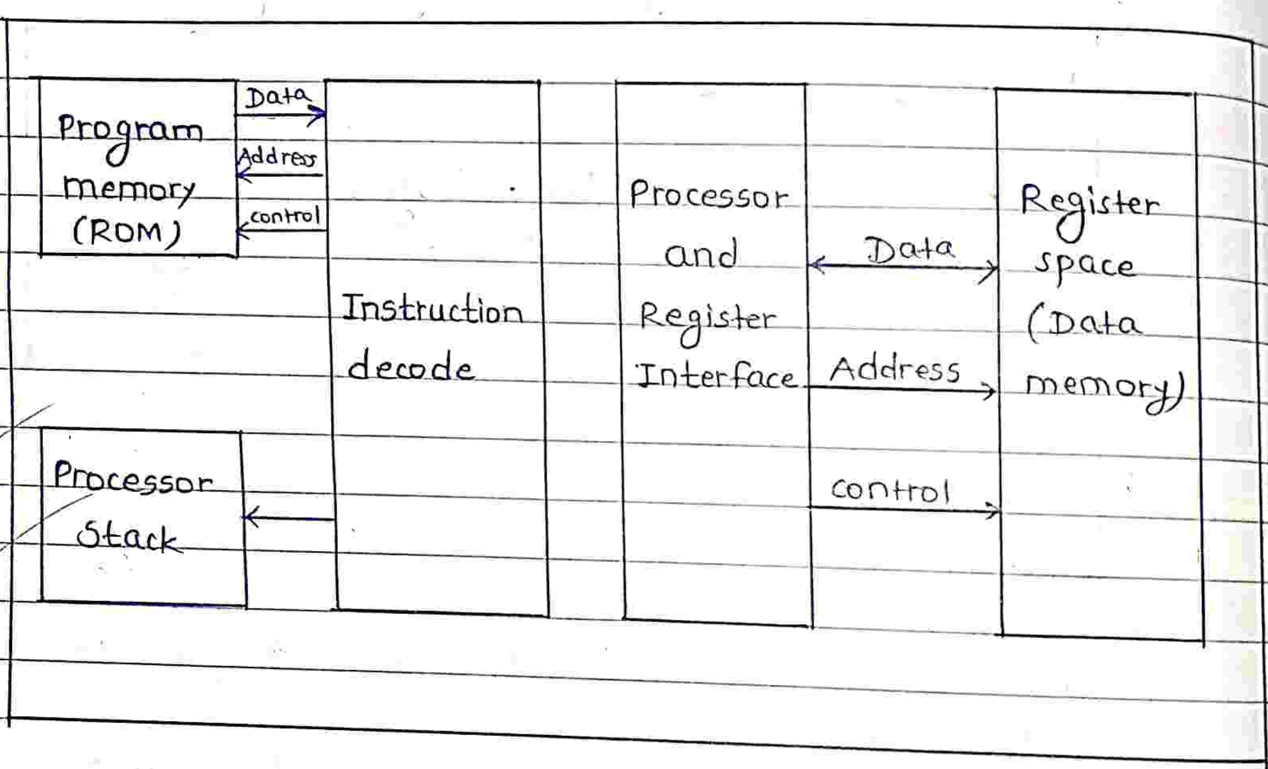
Que.7] Explain and draw basic structure of Harvard Architecture. State differences between Harvard & Von-Neumann Architecture.

⟹ Answer :

i) Harvard architecture provides seperate memory banks for program storage (code memory), the processor stack & variable RAM (data memory).

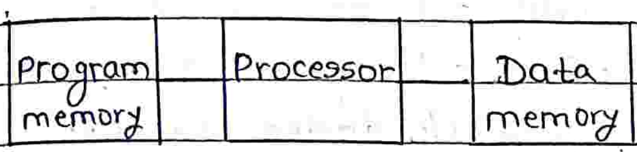
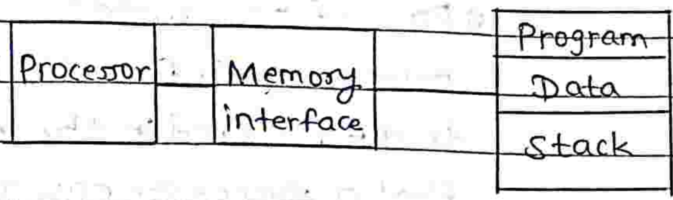
ii) It has advantage of executing instructions in fewer instructions cycles than Princeton (von Neumann) architecture.

iii) Greater amount of instruction parallelism can be achieved due to seperate memory banks.



iv) Comparison :

Harvard	Von-Neumann
i) It consists of seperate memory bank, processor, instruction decode.	i) It consist of processor, instruction decoder, memory interface unit.

Harvard	Von-Neumann
	
This architecture consists of seperate code (program) & data memory	This architecture consists of memory i.e. program (code) & data used same memory area.
ii) Execution of instruction is faster because fetching of code & data seperately or simultaneously.	ii) Execution of instruction is relatively slower than the Harvard Architecture because not possible to fetch code & data simultaneously.
iii) RISC architecture used only.	iii) RISC & CISC architecture used.
iv) More parallelism.	iv) Less parallelism compared to Harvard architecture.
v) e.g. Microcontroller (Real-Time system)	v) e.g. Microprocessor (general purpose).

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Que.1] What is meant by Addressing Mode? Explain all Addressing modes with example.

⇒ Answer:

i) Part of programming flexibility for each processor is the number & different kind of ways the programmer can refer to data stored in the memory or I/o device. The different ways that a processor can access data are referred to as addressing schemes or addressing modes.

ii) The different addressing modes are :-

1] Immediate 5] Register Indirect
2] Direct 6] Displacement (Indexed)
3] Indirect 7] Relative
4] Register 8] stack.

iii) Immediate Addressing Mode :-

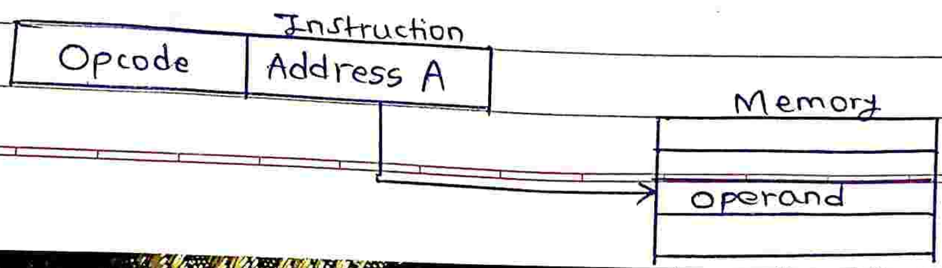
- In this case, operand is part of instruction. operand is in the immediate next location of opcode, hence the name.
- It is fast addressing mode, but has limited range.

Instruction

Opcode	Operand

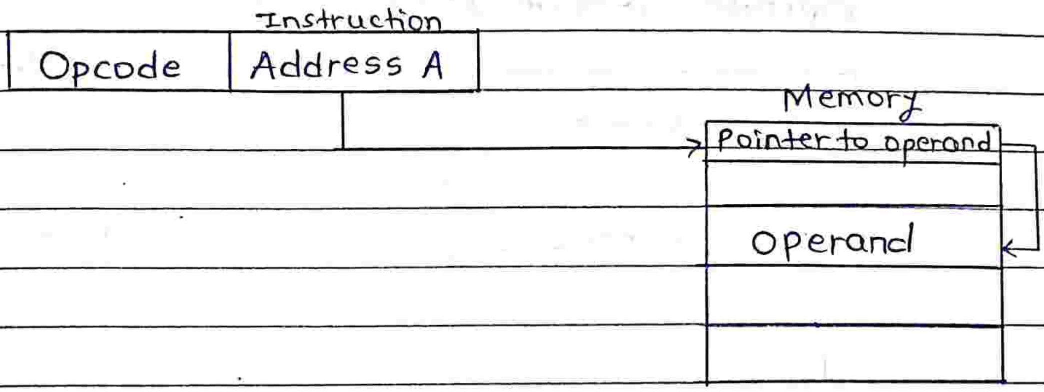
iv) Direct Addressing Mode :-

- In this case, address field contains memory address of the operand.
- In this case, there is only single memory reference to access data.
- Advantage is that there are no calculations to work out effective address.
- Disadvantage is that this mode can be used for a limited address space.



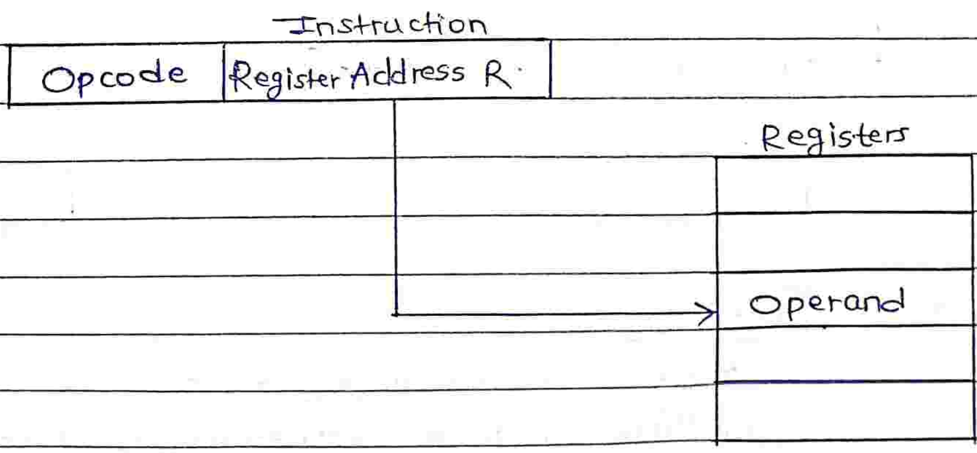
v) Indirect Addressing Mode :-

- In this case, memory location has the address of operand in another memory location i.e a memory operand is pointed to by address field contains address of (pointer to) operand.
- This method is slower as multiple memory locations are to be accessed to get operand.



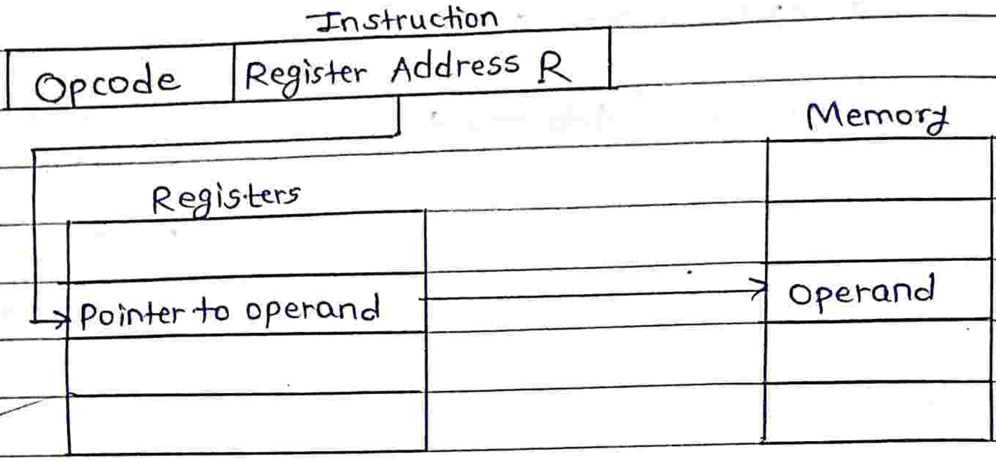
vi) Register Addressing Mode :-

- In this case, operand is held in the register named in operand address field.
- There are limited no. of registers; hence very small address field is required. Hence shorter & faster Instruction fetch.
- Advantage : no memory access.
- Disadvantage : Limited no. of registers available in most of the processors.
- very fast execution but limited address space.



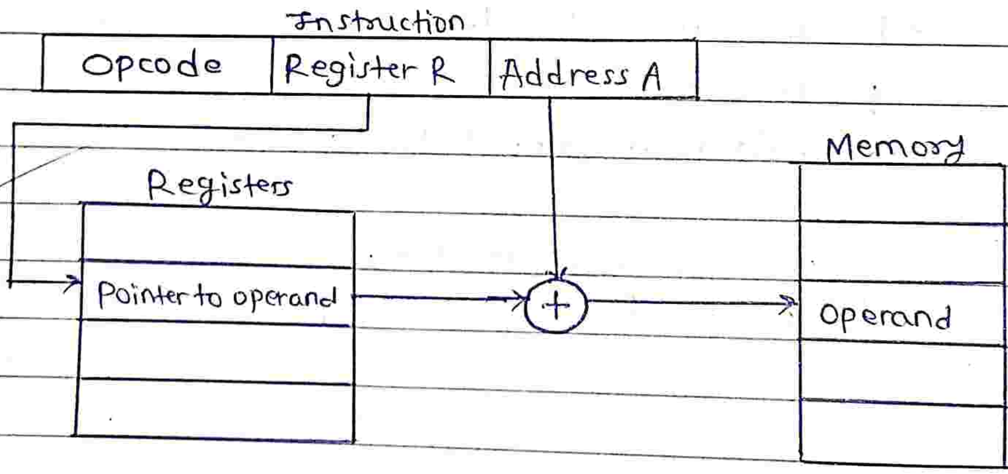
vii) Register Indirect addressing :-

- In this case, operand memory address is pointed to by contents of register R.
- It requires one less memory access than indirect addressing mode.



viii) Displacement addressing mode :-

- In this case, there are two address fields that hold base address & displacement.
- Advantage : Pointer need not be continuously updated.
- Disadvantage : Extra computation required for address calculation.



ix) Stack Addressing mode :-

- Used to access data from top of stack.
- It uses PUSH & POP instructions to access stack.
- e.g POP AX ; Pop two top items bytes from top of stack.

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Que.2] Differentiate between RISC & CISC Architecture.

⟹ Answer :

Sr. No.	Properties	RISC	CISC
1)	No. of instructions	Less	More
2)	Addressing Modes	Less	More
3)	Instruction Formats	Less	More
4)	Instruction Size	Fixed	variable
5)	control Unit	Hardwired	Micro-programmed
6)	No. of Bus cycles to execute instruction.	Single CPU cycle (for 80% instructions)	Multiple CPU cycles
7)	Control Logic & Decoding subsystem	Simple	Complex
8)	Pipelining	Huge no. of stages of pipelining.	Difficulty in efficient implementation
9)	HHL instructions	Supported	Not supported.

Que.3] What is Machine Instruction? Explain elements of machine instruction. Describe 0-1-2-3 address format of machine instruction.

⟹ Answer :

i) The operation of CPU and computer system is determined by instructions executed by CPU. These instructions are known as machine instructions.

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ii) They are in the form of binary codes.

iii) A particular sequence of these binary codes is used to perform particular task is known as machine language program.

iv) Each Instruction of CPU has specific info. fields which are required to execute it. These info. fields of machine instructions are called elements of machine instruction.

v) Elements of machine instructions are :-

① Operation code : Specifies the operation to be performed by binary code.

② Source/Destination operand : It specifies source/destination operand for instruction.

③ Source operand address : The operation specified by the instruction may require one/more source operands.

④ Destination operand address : The operation executed by CPU may produce result. Usually, result is stored in destination operand.

⑤ Next Instruction address : It tells the CPU, from where to fetch next instruction after completion of execution of current instruction.

vi) According to address references, there are four types of instruction formats :

① <u>Three address instructions</u> : It can be represented symbolically as ADD C, A, B where A, B, C are the variables. ($C \leftarrow A+B$) These variable names are assigned to distinct locations in the memory. In this instruction operands A&B are called source operands & operand C is called destination operand & ADD is the operation to be performed on operands.

② <u>Two address instructions</u> : It can be represented symbolically as ADD A, B. This instruction adds the contents of variables A&B & stores sum in variable A destroying its previous contents ($A \leftarrow A+B$). Here, B = source operand, A = source & destination operand

③ One Address Instruction : It can be represented symbolically as ADD B. This instruction adds the contents of variable B, into the processor register called accumulator (AC) & stores the sum back into the accumulator destroying the previous contents of accumulator ($AC \leftarrow AC+B$).

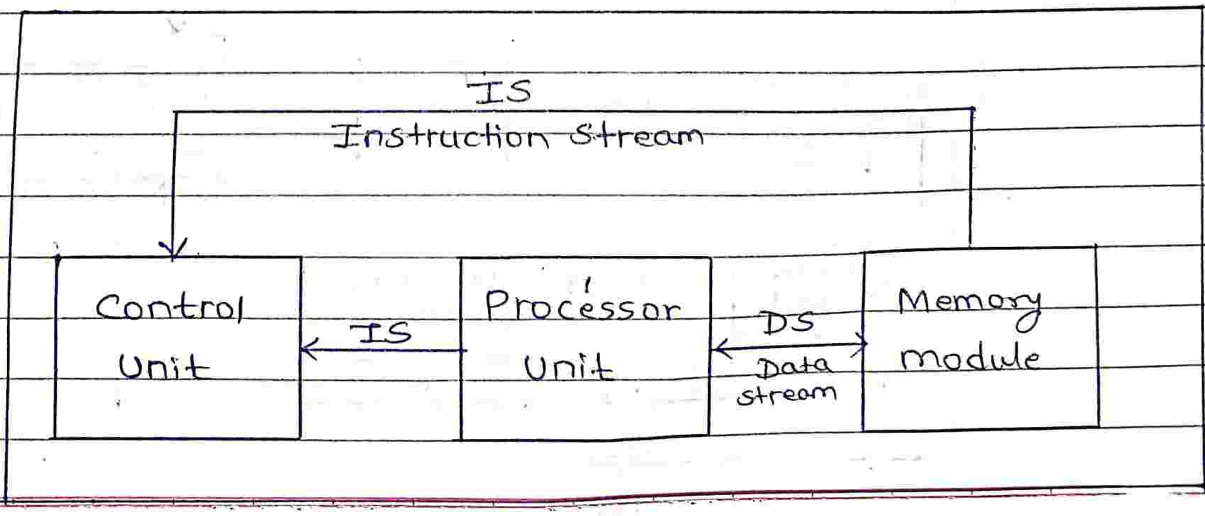
④ Zero Address Instruction : Locations of all operands are defined implicitly. e.g. CMA instruction complements contents of accumulator (AC) & stores complemented contents back into accumulator ($AC \leftarrow \overline{AC}$). Such instructions are also found in machines that store operands in structure called pushdown stack.

Que.4] Draw and Explain SISD, SIMD, MISD, MIMD Architecture.

⟹ Answer :

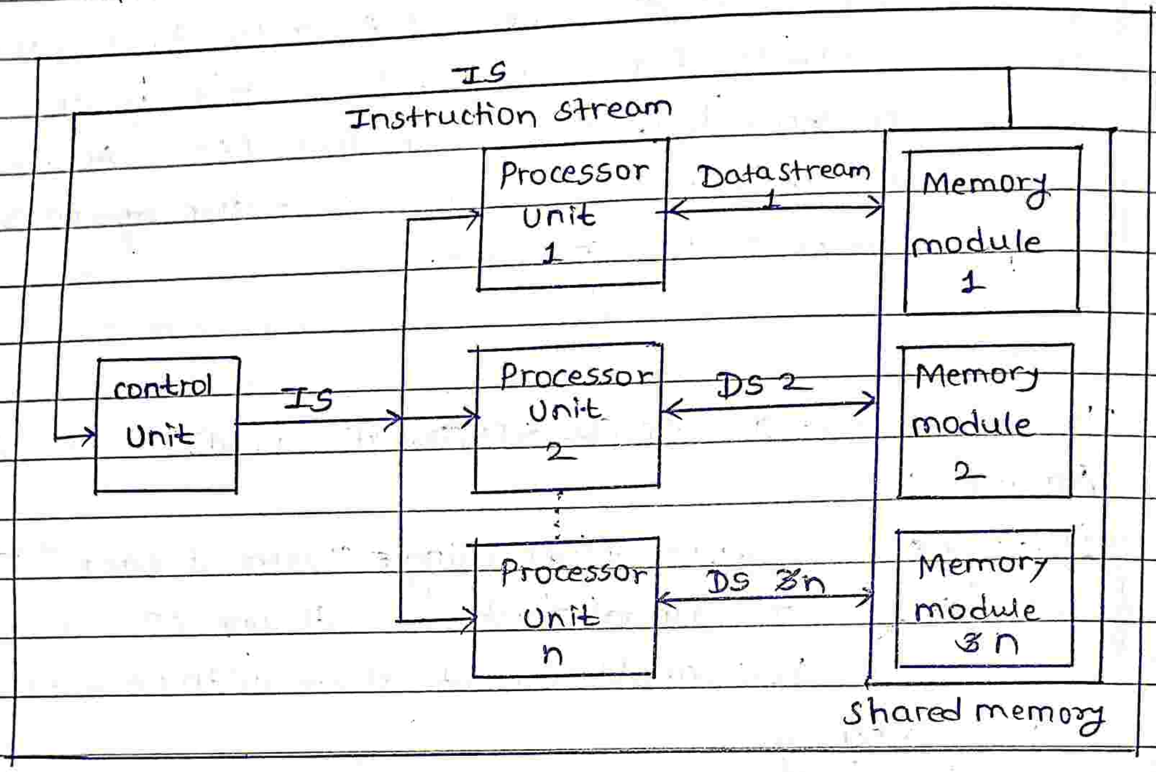
1] Single Instruction Stream - Single Data stream (SISD) :

- Most conventional machines with one CPU containing a single ALU capable only of scalar arithmetic fall into this category.
- SISD computers & sequential computers are thus synonymous.
- In SISD, computers instructions are executed sequentially but may overlap in their execution stages (Pipelining). They may have more than one functional unit, but all functional units are controlled by single control unit.

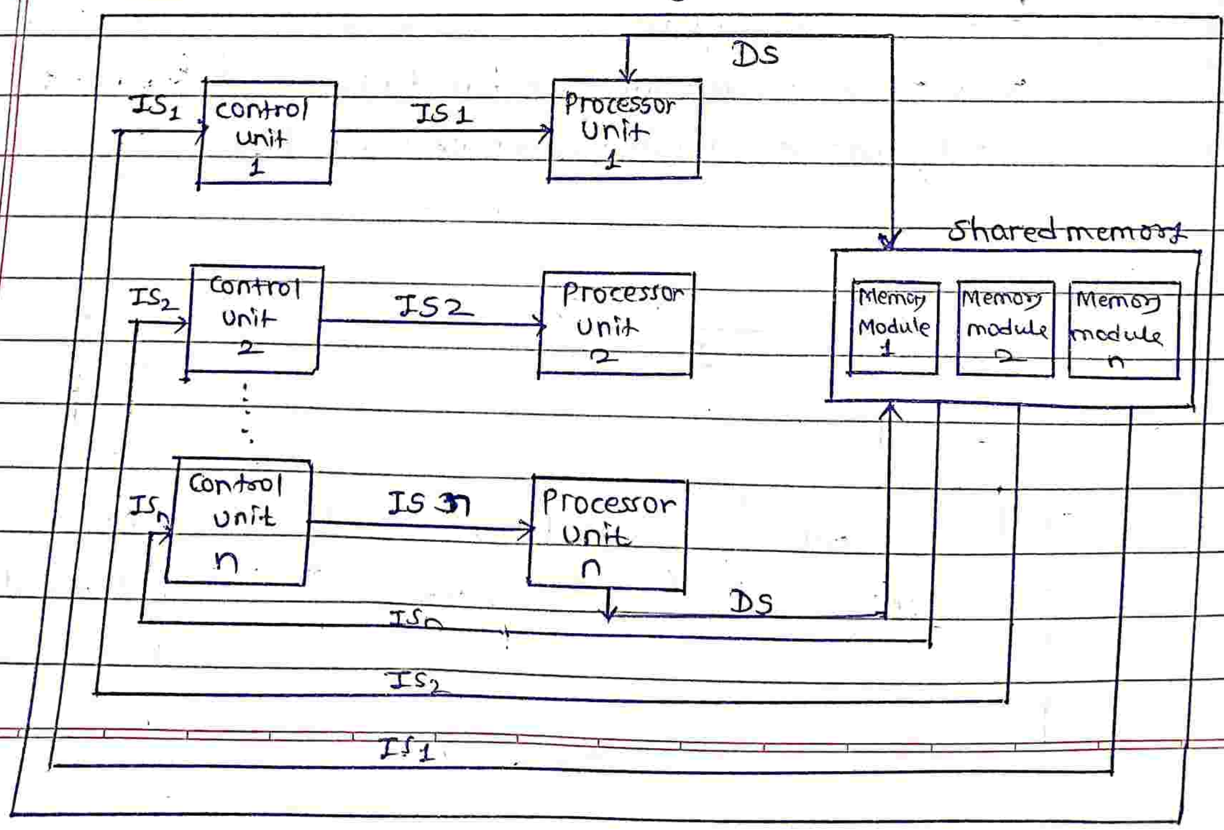


2] Single Instruction Stream Multiple Data Streams (SIMD):

- This corresponds to array processors.
- They have multiple processing/execution units & one control unit. ∴ All units are supervised by single control unit.
- Also known as single program, multiple data stream (SPMD).

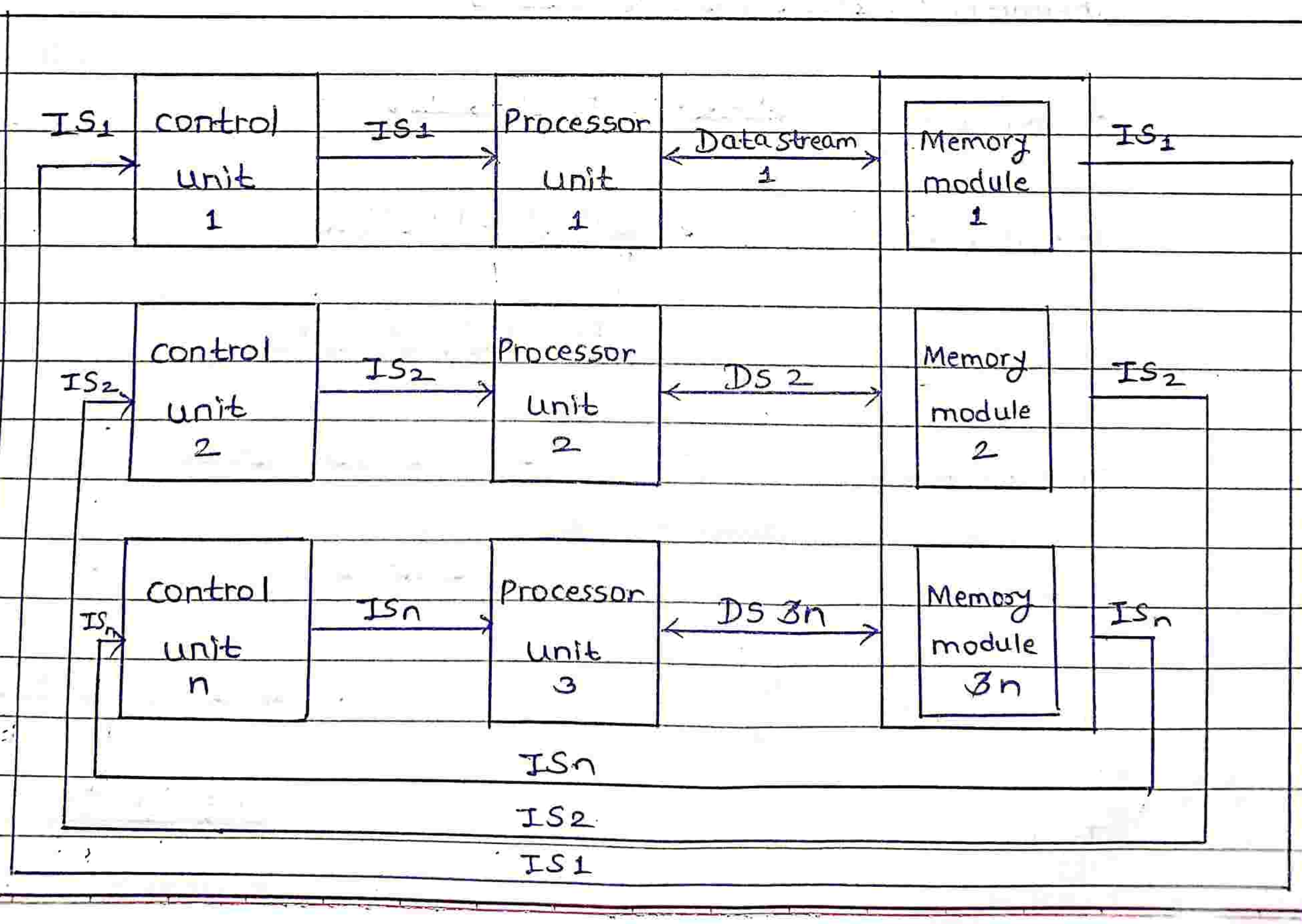


3] Multiple Instruction Stream Single Data Stream (MISD):



- Not many parallel processors fit well into this category. In MISD, there are n processor units. Each receiving distinct instructions operating over same data stream & its derivatives.
- Results of one processor become input of next processor in micropipe.

4] Multiple Instruction streams Multiple Data streams (MIMD) :-
- Most multiprocessors system & multiple computer system can be classified in this category.
- In MIMD, there are more than one processor unit having ability to execute several program simultaneously.
- MIMD computer implies interactions among the multiple processors because all memory streams are derived from same data space shared by all processors.
- If the n data streams are derived from disjointed subspaces of shared memories then we would have so-called Multiple SISD (MSISD) operation.



Que.5] Explain Instruction Pipelining w.r.t. operation & speed up.

⟹ Answer :

i) Various cycles involved in the instruction cycle. These fetch, decode & execute cycles for several instructions performed simultaneously to reduce overall processing time. This process is referred to as instruction pipelining.

ii) Let us assume that instruction execution is divided into following five stages :

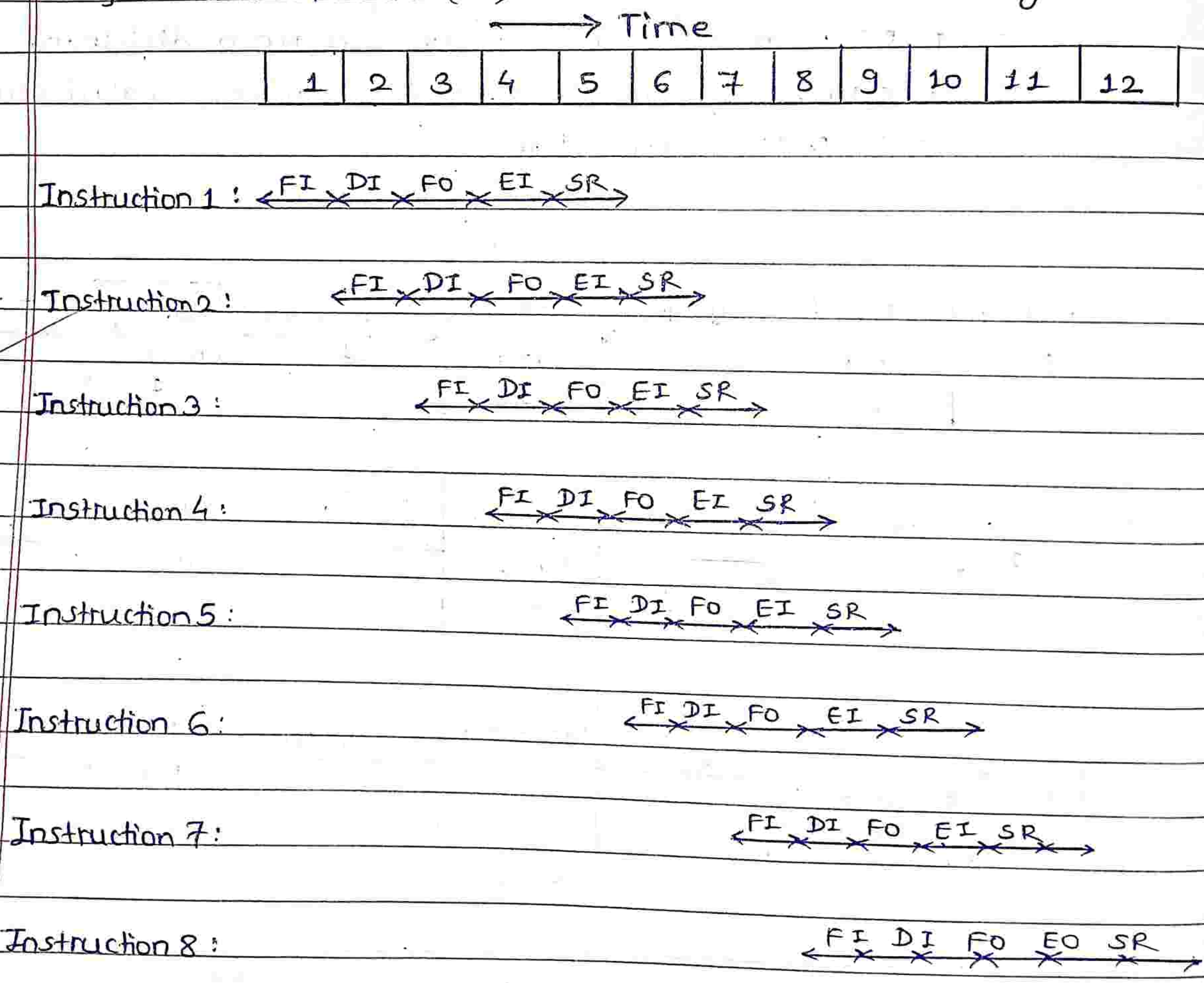
1] S_1 – Fetch Instruction (FI) : Read the next instruction into an instruction buffer.

2] S_2 – Decode Instruction (DI) : Decode the opcode.

3] S_3 – Fetch operands (FO) : Fetch each memory referenced operand.

4] S_4 – Execute Instruction (EI) : Perform the indicated operation.

5] S_5 – Store Results (SR) : Store the result in memory



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iii) With this subdivision & assuming equal duration for each stage, we can reduce the execution time for 8 instructions from 40 time units to 12 time units.

iv) A Pipeline's performance can be measured by its throughput in terms of millions of instructions executed per second i.e. MIPS.

v) Another popular measure of performance is no. of clock cycles per instruction i.e. CPI.

$$\boxed{CPI = f/MIPS}$$

where, f = pipeline's clock frequency in MHz.

Values of CPI & MIPS are avg. figures.

vi) Another general measure is speedup S(m) defined by,

$$\boxed{S(m) = \frac{T(1)}{T(m)}}$$

where, T(m) = execution time for some target program on m-stage line.
T(1) = execution time for same program on similar nonpipelined processor.

Que.6] Explain Interrupt w.r.t. its purpose, types. Describe step by step Interrupt handling procedure of microprocessor.

⟹ Answer:

i) Sometimes it is necessary to have the computer automatically execute one of a collection of special routines whenever certain conditions exists within program or computer system. e.g. It is necessary that computer system should give response to devices such as keyboard, sensor & other components when they request for a service. The purpose of interrupt is to provide above functionality.

ii) Interrupt provides an external asynchronous input that would inform the processor that it should complete whatever instruction that is currently being executed & fetch a new routine that will service requesting device.

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iii) There are two main classes of Interrupts:
- Hardware Interrupts.
- Software Interrupts.

iv) An Interrupt caused by external signal is called as hardware interrupt & Interrupt caused by special instructions are called software interrupts.

Que.7] What is meant by multi-core architecture? What is its advantage? List typical features of multicore intel core i7.

⟹ Answer:

i) Multicore architecture places multiple processor cores & bundles them as a single physical processor.

ii) The objective is to create a system that can complete more tasks at same time, therby gaining better overall system performance.

iii) Features of Intel core i7 are :-

1] Intel core i7 is a high end processor. It has higher clock speed, bigger cache & hyper threading features.

2] i7 mobile processors are quad-core.

3] i7 processors support/come with Ivy bridge.

4] i7 processors come with direct media interface and integrated GPU.

5] i7 processors come with turbo boost facility.

6] It has 32 kB Instruction & 32 kB data first level cache (L1) for each core.

7] It can have upto 8 MB of shared instruction/data third level cache (L3), shared among all cores.

8] It supports 1/2 channels of DDR3 memory.

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Que.1] Compare SRAM and DRAM.

⟹ Answer :

Static RAM	Dynamic RAM
i) Static RAM contains less memory cells per unit area.	i) Dynamic RAM contains more memory cells as compared to SRAM per unit area.
ii) It has less access time hence faster memories.	ii) Its access time is greater than SRAM.
iii) Static RAM consists of a number of flip-flops. Each flip-flop stores one bit.	iii) Dynamic RAM stores the data as a charge on the capacitor. It consists of MOSFET & capacitor for each cell.
iv) Refreshing circuitry is not required.	iv) Refreshing circuitry is required to maintain the charge on the capacitors after every few milliseconds.
v) cost is more	v) cost is less.

* Compare Memory mapped I/O and I/O mapped I/O.

⟹ Answer :

Memory mapped I/O	I/O mapped I/O
i) Memory and I/O Shared the entire address range of processor.	i) Processor provides seperate address range for memory & I/O devices.

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Memory mapped I/O	I/O mapped I/O
ii) Usually, processor provides more address lines for accessing memory. Therefore, more decoding is required control signals	ii) Usually, processor provides less address lines for accessing I/O. Therefore, less decoding is required.
iii) Memory control signals are used to control read and write I/O operations.	iii) I/O control signals are used to control read & write I/O operations.

Que.2] Along with diagram, explain Fully associate cache mapping technique.

⟹ Answer:

i) Following figure shows fully associative mapping technique. In this technique, a main memory block can be placed into any cache block position. As there is no fix block, the memory address has only two fields: word & tag.

ii) This technique gives complete freedom in choosing the cache location in which to place the memory block. Thus, the memory space in the cache can be used more efficiently.

iii) A new block that has to be loaded into the cache has to replace (remove) an existing block only if cache is full.

iv) In such situations, it is necessary to use one of the possible replacement algorithm to select block to be replaced.

v) Disadvantage: In associative-mapped cache, it is necessary to compare higher order bits of address of main memory within all tags corresponding to each block to determine whether given block is in cache.

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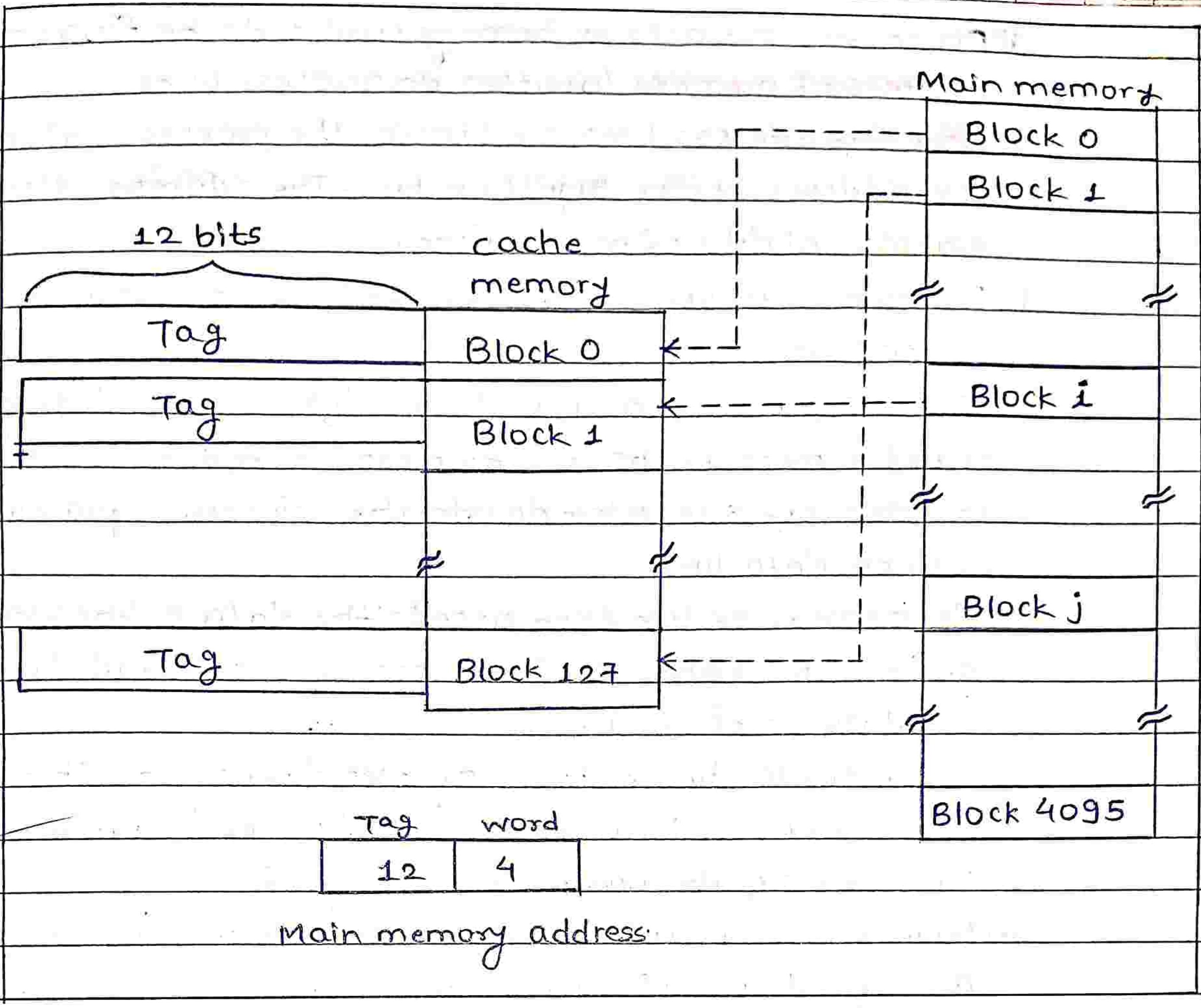
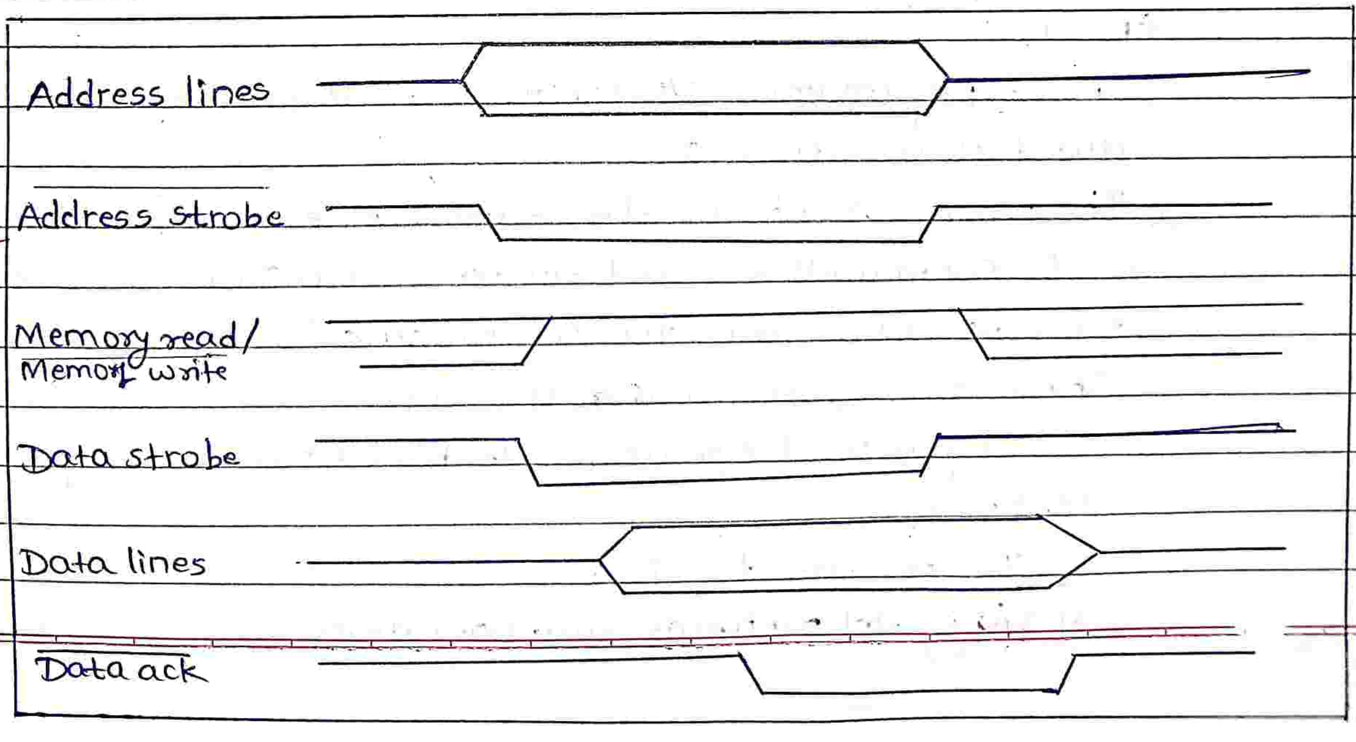


Que.3] Explain memory Read cycle with help of suitable timing diagram.

⟹ <u>Answer</u>:



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i) Processor initiates a memory read cycle by floating the address of memory location on address lines.
ii) Once the address lines are stable, the processor asserts the address strobe signal on bus. The address strobe signals validity of address lines.
iii) Processor then sets the memory Read/write signal to high i.e. read cycle.
iv) Now the processor asserts data strobe signal. This signals to be the memory that the processor is ready to read data.
v) The memory subsystem decodes the address & places the data on data lines.
vi) The memory system then asserts the data acknowledge signal. This signals to the processor that valid data is available on data bus.
vii) Processor latches in data & negates data strobe. This signals to memory that the data has been latched by processor.
viii) Processor negates address strobe signal.
ix) Memory system then negates data acknowledgement signal. This signals end of memory read cycle

Que.4] Explain program controlled I/O with help of flowchart.
⇒ Answer:
i) In the programmed I/O method of interfacing, CPU has direct control over I/O.
ii) The processor checks the status of devices & issues read or write commands & then transfers data. During the data transfer, CPU waits for I/O module to complete operation & hence this system wastes CPU time.
iii) The sequence of operations to be carried out in programmed I/O operation are:
1] CPU requests for I/O operation.
2] I/O model performs said operation.

3] I/O module updates status bits.

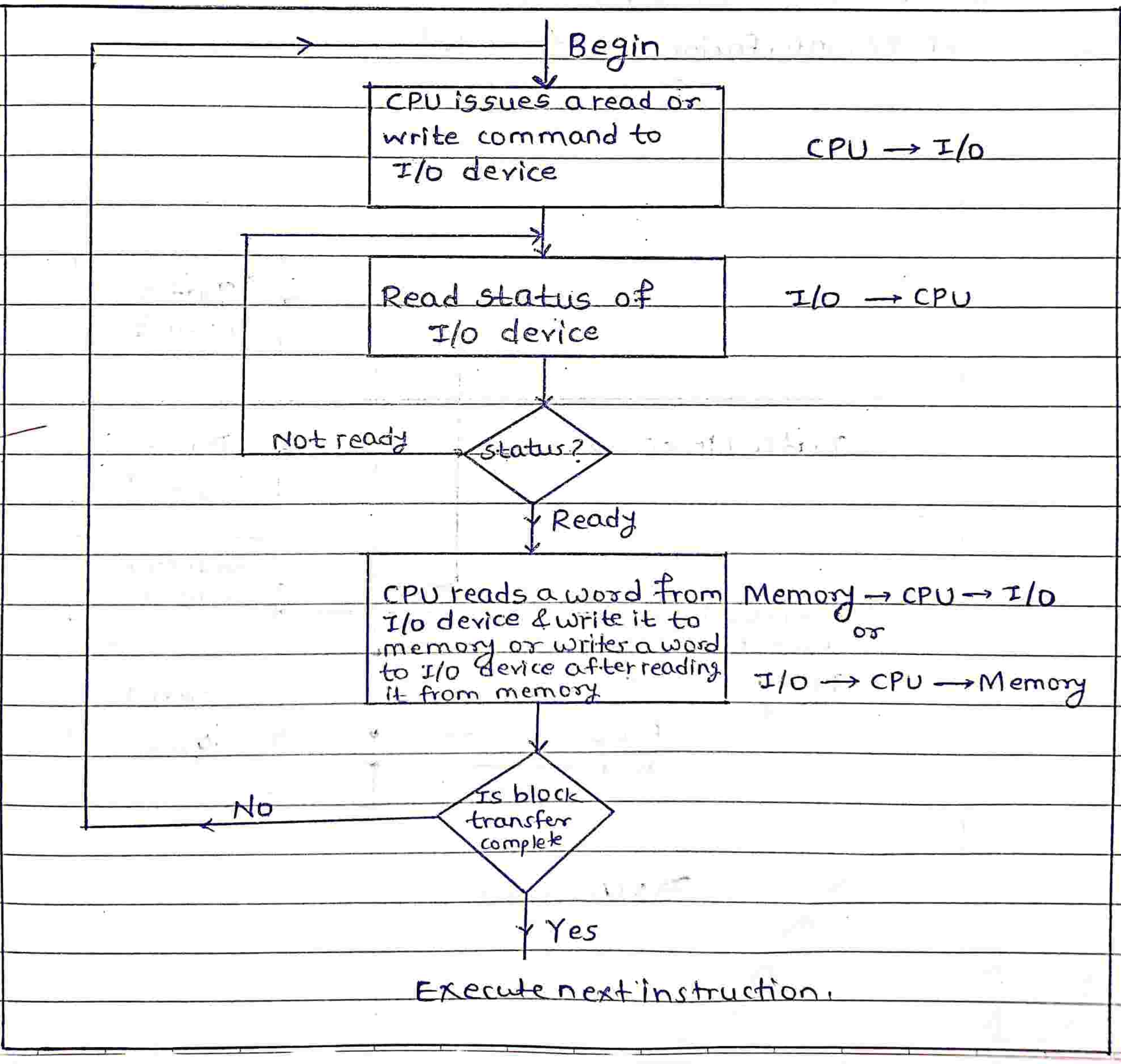
4] CPU checks these status bits periodically. Neither the I/O module cannot inform CPU directly nor can I/O module interrupt CPU.

5] CPU may wait for the operation to complete or may continue operation later.

iv) IC 8255 is generally used as I/O module for programmed I/O method of interfacing.

v) A common programming task is the transfer of a block of words between an I/O device & memory.

vi) Given flowchart shows transferring a block of data.



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Que.5] What is DMA ? Along with suitable diagram, explain how DMA is used for data transfer.

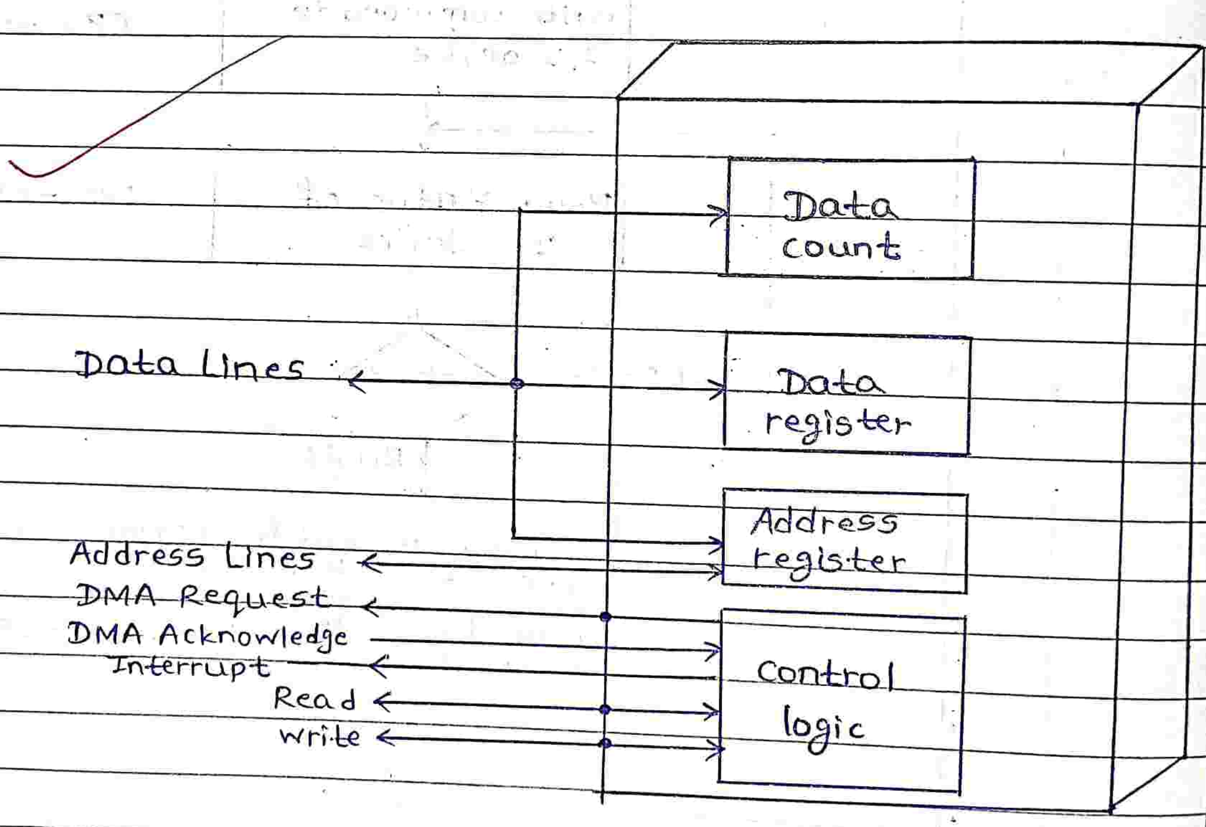
⟹ Answer :

i) DMA stands for Direct Memory Access. The I/O module can directly access (read or write) the memory using this method.

ii) Interrupt driven & programmed I/O require active operation of CPU, hence transfer rate is limited and CPU is also busy doing the transfer operation. DMA is the solution for these problems.

iii) DMA controller takes over the control of the bus for CPU from for I/O transfer.

iv) The internal block diagram of DMA controller of I/O module of I/O interfacing is shown below:



DMA controller

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v) There are various modes of operation used to transfer the data between the memory and I/O device by DMA controller.

vi) Four major methods used are :

1] Single Transfer mode.

2] Block Transfer mode.

3] Demand Transfor mode.

4] Hidden Transfer mode.

vii) Advantage of single Transfer mode is that CPU has not to remain out of the system or not having the access of the system bus for longer time, instead only for one transfer.

viii) In Block Transfer mode, I/O device gets the transfer of data at very faster speed.

ix) In Demand Transfer mode, device continues making transfers until Terminal court is encountered or until DREQ goes inactive.

x) In Hidden Transfer mode, DMA controller takes over the charge on the system bus & transfers data when processor does not needs system bus. Processor does not even realize of this transfer being taken place.

22/11/23